

Get the Data

- As before, download US State data for Single Family Homes from Zillow:
<https://www.zillow.com/research/data/>

HOME VALUES

Zillow Home Value Index (ZHVI): A measure of the typical home value and market changes across a given region and housing type. It reflects the typical value for homes in the 35th to 65th percentile range. Available as a smoothed, seasonally adjusted measure and as a raw measure.

Zillow publishes top-tier ZHVI (\$, typical value for homes within the 65th to 95th percentile range for a given region) and bottom-tier ZHVI (\$, typical value for homes within the 5th to 35th percentile range for a given region).

Zillow also publishes ZHVI for all single-family residences (\$, typical value for all single-family homes in a given region), for condo/coops (\$, for all homes with 1, 2, 3, 4 and 5+ bedrooms (\$), and the ZHVI per square foot (\$, typical value of all homes per square foot calculated by taking the estimated home value for each home in a given region and dividing it by the home's square footage).

Note: Starting with the January 2023 data release, and for all subsequent releases, the full ZHVI time series has been upgraded to harness the power of the neural Zestimate.

More information about what ZHVI is and how it's calculated is available on [this overview page](#). Here's a handy [ZHVI User Guide](#) for information about properly citing and making calculations with this metric.

Data Type

ZHVI Single-Family Homes Time Series (\$)

Geography

State

Download

- Compile the year-end data (12/31/20yy) dat for **your assigned state** for each of the past **10** years and format it like this in an Excel spreadsheet:

	A	B
1	YearEndData	AvgHomeValue
2	12/31/2015	\$163,202.97
3	12/31/2016	\$171,967.12
4	12/31/2017	\$182,124.96
5	12/31/2018	\$192,940.13
6	12/31/2019	\$205,353.77
7	12/31/2020	\$226,173.40
8	12/31/2021	\$250,635.68
9	12/31/2022	\$269,983.34
10	12/31/2023	\$287,998.06
11	12/31/2024	\$304,518.03

Data Analysis

Using this data, do the following in Pandas:

- Read this data into a **DataFrame** object named **xxTenYearEndDF** (where "**xx**" is the two-letter abbreviation for your assigned state).
- Then, using **subsetting**, pull out the column that contains the average home values and assign it to a series named **xxYearEndDatesSeries** (where "**xx**" is the two-letter abbreviation for your assigned state).
- In the console, display "**Year End Averages:**".
- In the console, print **xxYearEndDatesSeries**
- In the console, display "**Third year:**" followed by the year-end average home value for the third year. For this, use **iloc[]** notation.
- In the console, display "**Sixth year:**" followed by the year-end average home value for the sixth year. For this, use **iloc[]** notation.
- In the console, display "**Ninth year:**" followed by the year-end average home value for the ninth year. For this, use **iloc[]** notation.
- Write Pandas code to assign **new row** labels to the DataFrame. The row labels should have the following format: **ssyearEnd20yy**
 - **ss** is the two-letter abbreviation for your assigned state.
 - **yy** is the last two digits of that year

Example for Wisconsin:

	YearEndData	AvgHomeValue
WIyearEnd2014	12/31/2014	\$154,512.98
WIyearEnd2015	12/31/2015	\$161,577.39
WIyearEnd2016	12/31/2016	\$170,254.24
WIyearEnd2017	12/31/2017	\$180,310.90
WIyearEnd2018	12/31/2018	\$191,018.35
WIyearEnd2019	12/31/2019	\$203,308.35
WIyearEnd2020	12/31/2020	\$223,920.60
WIyearEnd2021	12/31/2021	\$248,139.22
WIyearEnd2022	12/31/2022	\$267,294.16
WIyearEnd2023	12/31/2023	\$285,129.45

- Then, write Pandas code that uses **loc[]** to display the **first**, **fifth**, and **last** rows using the **new row labels**. Your output should be something like this:

```
First row:
YearEndData      12/31/2014
AvgHomeValue     $154,512.98
Name: WIyearEnd2014, dtype: object

Fifth row:
YearEndData      12/31/2018
AvgHomeValue     $191,018.35
Name: WIyearEnd2018, dtype: object

Last row:
YearEndData      12/31/2023
AvgHomeValue     $285,129.45
Name: WIyearEnd2023, dtype: object
```

Submit Your Code and Results

- Copy and paste your **Thonny code** into a Word document
- Copy and paste your **Console output** into the same Word document
- **Save** this Word file as a **PDF**.
- **Submit** this PDF via **Canvas**.